# INF6422E Advanced Concepts in Computer Security

# Practical Work 3 – Winter 2026

# Collaborative and Privacy-Aware Anomaly-Based IDS using Isolation Forest

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## Instructions:

- This work is to be submitted in a group of two or three and via Moodle only.
- The submitted report must be in pdf form and also include (.py, .ipynb) with it. You're free to build it in any format you want, however (.docx, .odt, .tex, etc.).
- The report must contain a title page including the course title, the lab title, your names and student ID numbers (Matricule).
- The report must be submitted by the 1st of March 2026 before 23h59. A penalty of 10% will be applied for each day after that date.
- There are 4 labs in total, each of 20 marks/10 weightage.

## Objective:

This lab studies trustworthy and privacy-aware intrusion detection in collaborative environments using an anomaly-based IDS. Unlike Practical Work 1 (supervised IDS classification) and Practical Work 2 (security of ML under adversarial manipulation), Practical Work 3 focuses on deployment-oriented challenges:

- How to detect attacks when only benign traffic is used for training (anomaly-based IDS),
- How collaboration among multiple IDS nodes changes detection outcomes,
- How privacy constraints (limited sharing) affect performance,
- How to evaluate trade-offs using quantitative security metrics.

# Key Learning Goals:

1. Understand anomaly-based intrusion detection using benign-only training in realistic IDS deployments.
2. Implement centralized and collaborative IDS architectures using Isolation Forest models.
3. Evaluate the impact of privacy constraints through limited feature sharing in collaborative IDS.
4. Analyze trade-offs between detection performance, privacy protection, and trust assumptions.
5. Introduce confidence-based decision interpretation to support trustworthy IDS operation without overlapping advanced semantic security topics.

# Dataset

Use the same CICIDS2017 configuration assigned in Practical Work 1. Monday (benign baseline) + one selected attack day (Tuesday/Wednesday/Thursday/Friday).
Important rule for Lab 3:

- Training for anomaly detection must use Monday (benign) only.
- Attacks appear only at test time (your assigned day + optional benign from Monday).

**Dataset Link:** https://www.unb.ca/cic/datasets/ids-2017.html

or

https://drive.google.com/drive/folders/1nbGBRm1rsxYvdHYNUf8ovJPiELUeeCp_?usp=sharing

# 1. Centralized Anomaly-Based IDS with Isolation Forest [6 points]

### 1.1 Build the Centralized Anomaly IDS

Train an Isolation Forest model using only benign traffic (Monday).

- Use Isolation Forest from scikit-learn
- Use a fixed random seed
- You must state your contamination setting and justify it

**Deliverable:**

Describe the training setup (what data used, why only benign, parameter choices).

Report detection metrics on the test set (see Section 1.3).

### 1.2 Map Anomaly Outputs to IDS Labels

Isolation Forest produces binary predictions indicating whether a network flow is normal or anomalous:

- +1 → normal behavior
- -1 → anomalous behavior

In this lab, these outputs are mapped to IDS decisions as follows:

- anomaly (-1) → attack
- normal (+1) → benign

This mapping allows the anomaly-based model to function as an Intrusion Detection System, where deviations from learned benign behavior are treated as potential attacks.

**Deliverable:**

One paragraph explaining why anomaly detection is appropriate when attack labels are limited or unreliable.

### 1.3 Quantitative Performance Evaluation (Security Metrics)

Evaluate your centralized IDS on a test set containing:

- benign samples
- attack samples (from your assigned day)

Report the following:

- Confusion Matrix
- Precision, Recall, F1-score
- False Positive Rate (FPR)
- False Negative Rate (FNR)

**Deliverable:**

A results table + a short security interpretation:

- Which error is more dangerous for IDS: FPR or FNR (and why)?
- Does your anomaly IDS favour one type of error?

## 2. Collaborative IDS Simulation (3 Nodes/Models) [8 points]

### 2.1 Create Three IDS Nodes (Distributed Training)

In real-world networks, intrusion detection is often performed by multiple IDS sensors deployed in different departments or network segments. In this task, you will simulate such a distributed IDS environment.
Using only the benign traffic from Monday, divide the dataset into three non-overlapping subsets, each representing a separate IDS node:

- Node A: Benign subset A
- Node B: Benign subset B
- Node C: Benign subset C

Each subset must contain different samples, and no data point may appear in more than one subset. Train one Isolation Forest model per node, using only the benign traffic assigned to that node. As a result, you will obtain three independent IDS models, each learning a slightly different definition of "normal" network behavior.

**Deliverable:**

For this task, you must provide:
A clear explanation of how you partitioned the benign Monday data, for example:

- a random split,
- a time-based split, or
- a host-based split (if relevant features are available).

A performance evaluation for each IDS node, where:

- All three models are tested independently,
- The same test dataset (benign + attack traffic) is used for all nodes.

Report standard IDS metrics (e.g., precision, recall, F1-score, false positive rate) for Node A, Node B, and Node C separately.

**2.2 Collaborative Decision Rules (Fusion Strategies)**

Each IDS node trained with Isolation Forest supports two types of outputs:

- IDS labels (attack / benign), obtained using the mapping defined in Section 1.2
- Anomaly scores, obtained from the model's decision function

Depending on the assigned dataset, collaborative detection is performed using one of the following fusion strategies.

**Fusion Method Assignment**

| Dataset Configuration | Required Fusion Method |
|---|---|
| Monday + Tuesday | Majority Voting (IDS labels) |
| Monday + Wednesday | Majority Voting (IDS labels) |
| Monday + Thursday | Score Averaging (anomaly scores) |
| Monday + Friday | Score Averaging (anomaly scores) |

Each group must implement **only the method assigned** to its dataset.

**Method 1: Majority Voting (IDS Labels)**

In this method:

1. Each IDS node produces a binary IDS decision (attack or benign) using the mapping defined in Section 1.2.
2. The final collaborative decision is determined by majority vote across the three nodes.
   This approach reflects practical IDS deployments where sensors share alerts only, rather than detailed confidence information.

**Method 2: Score Averaging (Anomaly Scores)**

In this method:

1. Each IDS node outputs an anomaly score using the model's decision function.
2. Scores are normalized across nodes.
3. The normalized scores are averaged.
4. A threshold is applied to the averaged score to produce the final IDS decision.
   This approach preserves more information about detection confidence and uncertainty, which can be beneficial for complex attack scenarios.

**Deliverable:**

Comparison table:

- Centralized Isolation Forest
- Node A, B, C (standalone)
- Collaborative majority voting/Score Averaging

Short discussion:

- When collaboration improves detection
- When it increases false alarms

# 3. Privacy Protection in Collaborative IDS (Limited Sharing) [6 points]

In real organizations, IDS nodes often cannot share all traffic features due to privacy regulations, confidentiality concerns, or organizational policies. As a result, collaborative IDS systems must operate with limited shared information.
In this section, you will simulate privacy constraints by restricting the features used for training and collaboration.

### 3.1 Feature Sharing Restriction

You will evaluate collaborative IDS performance under two feature-sharing scenarios.

**Mode A: Full Feature Sharing (Reference Scenario)**

All IDS nodes are trained using the full set of features, as in Section 2. This mode represents an ideal scenario with no privacy restrictions.

**Mode B: Reduced Feature Sharing (Privacy-Aware Scenario)**

Limit the feature set used by all IDS nodes to simulate privacy protection. You may choose one of the following approaches:

- Top-k features based on prior analysis (from Practical Work 1), or
- A fixed small subset of features (e.g., 10–15 features) selected with justification (e.g., traffic volume, timing, rate-based features).

The same reduced feature set must be used by all IDS nodes in this mode.

**Deliverable:**

List the selected features used in Mode B.

Briefly justify:

- Why these features are less privacy-sensitive, and
- How reducing shared features helps protect privacy in collaborative IDS environments.

### 3.2 Quantifying the Privacy–Detection Trade-off

Compare collaborative IDS performance under:

- Mode A (full feature sharing), and

- Mode B (reduced feature sharing).

Evaluate how privacy constraints affect detection performance.

**Deliverable:**

Provide a performance comparison table showing the change in:

- Recall / False Negative Rate (missed attacks)
- False Positive Rate (false alarms)

Then answer the following questions:

- Which attack types (in your assigned day) are affected first when features are restricted?
- Why does limiting features reduce detection capability in anomaly-based IDS?

## Tools

Python + scikit-learn (required). Any plotting library is allowed.
No specialized privacy libraries are required.

## Resources:

https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html

https://scikit-learn.org/stable/auto_examples/ensemble/plot_isolation_forest.html

https://github.com/scikit-learn/scikit-learn/blob/main/sklearn/ensemble/_iforest.py

https://github.com/KianoushAmirpour/Intrusion_Detection_with_Unsupervised_Learning

https://github.com/youngdataspace/Detect-Outliers-Using-Isolation-Forest

## References:

1. Ripan, R. C., Islam, M. M., Alqahtani, H., & Sarker, I. H. (2022). Effectively predicting cyber-attacks through isolation forest learning-based outlier detection. *Security and Privacy*, *5*(3), e212. https://doi.org/10.1002/spy2.212
2. AbuAlghanam, O., Alazzam, H., Alhenawi, E. A., Qatawneh, M., & Adwan, O. (2023). Fusion-based anomaly detection system using modified isolation forest for internet of

things. *Journal of Ambient Intelligence and Humanized Computing*, *14*(1), 131-145. https://doi.org/10.1007/s12652-022-04393-9

3. Lifandali, O., Abghour, N., & Chiba, Z. (2023). Feature selection using a combination of ant colony optimization and random forest algorithms applied to isolation forest based intrusion detection system. *Procedia Computer Science*, *220*, 796-805. https://doi.org/10.1016/j.procs.2023.03.106